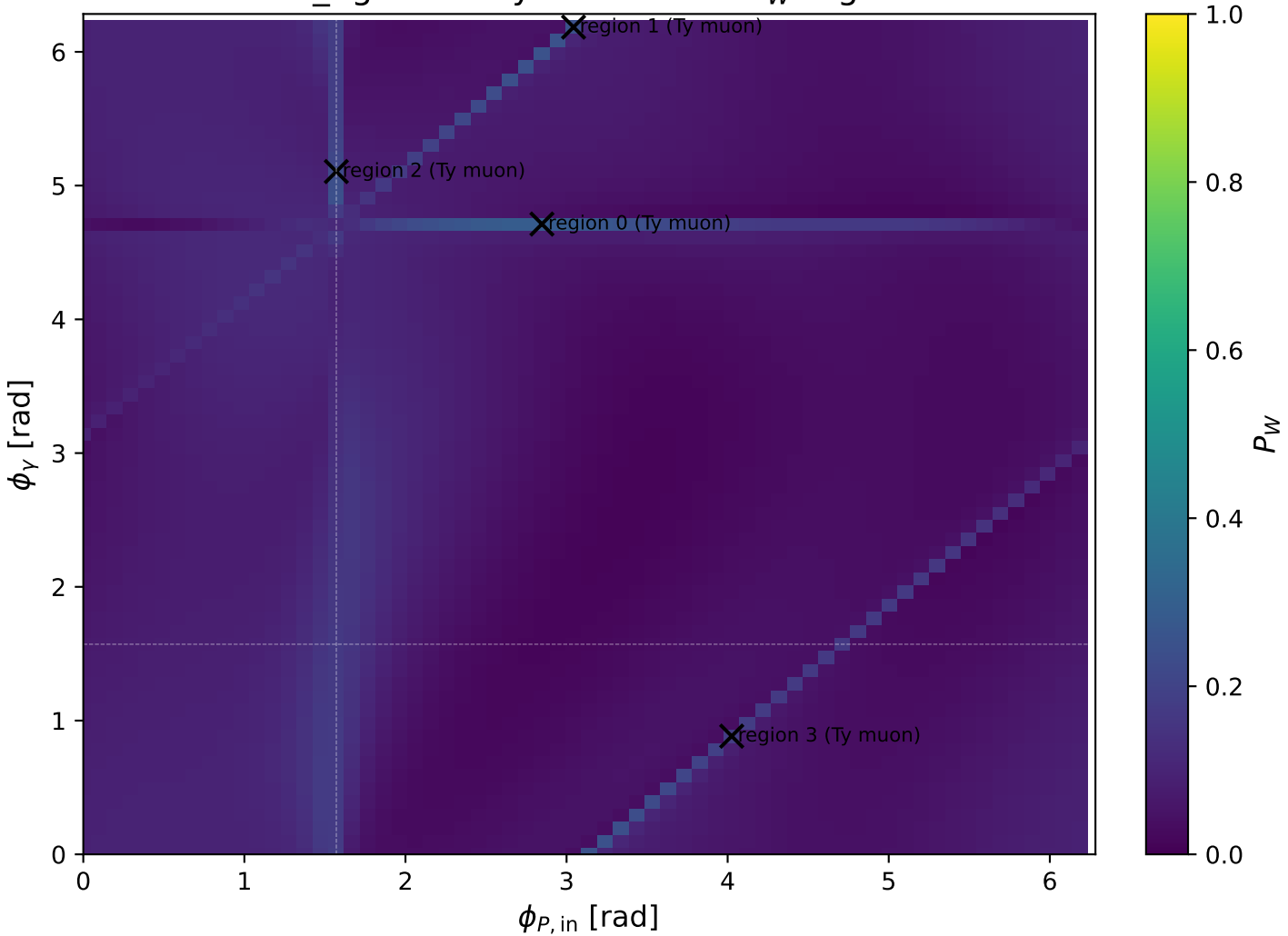
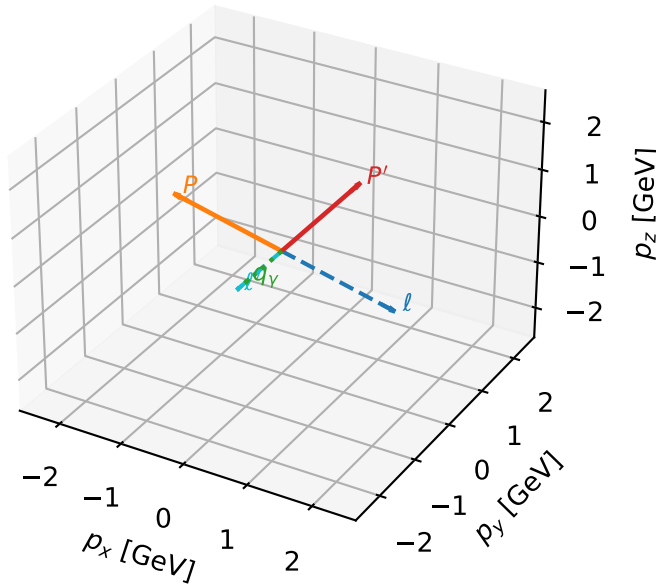


mid_Egamma Ty muon: max P_W regions



Ty muon characteristic max P_W configuration

3D momenta



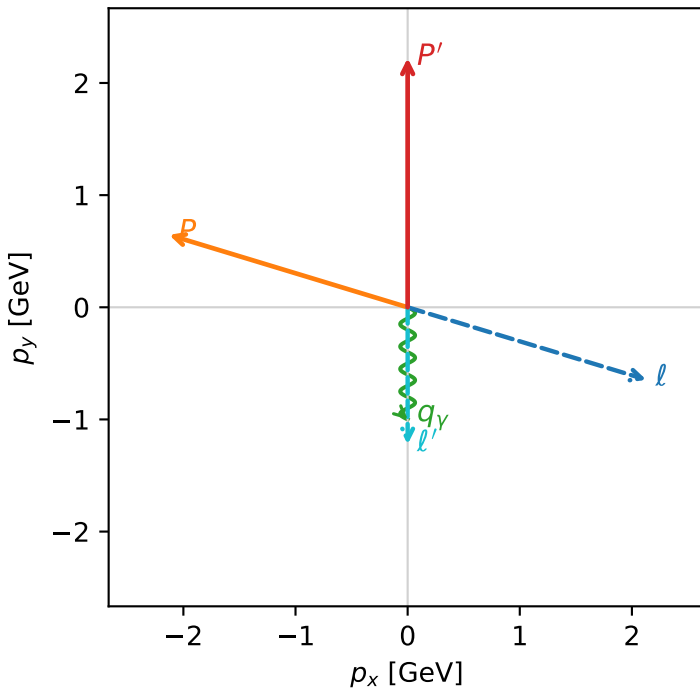
w_purity_mid_egamma_ty_lepton_region_0 $P_W=0.288889$
 region: mid_Egamma_Ty_lepton_region_0
 outgoing-state purity: 0.500411
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{y,\mu},h_p)$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=2.22204$
 $\theta_{in}=1.57079$, $\phi_l=5.98866$, $\phi_p=2.84707$
 $E_\gamma=1$, $\phi_\gamma=4.71239$
 $Q^2=3.86033$, $x_B=0.294047$, $t=-7.01178$, $W^2=10.1478$, $y=0.636526$
 $\theta(l',\gamma)=0$ deg

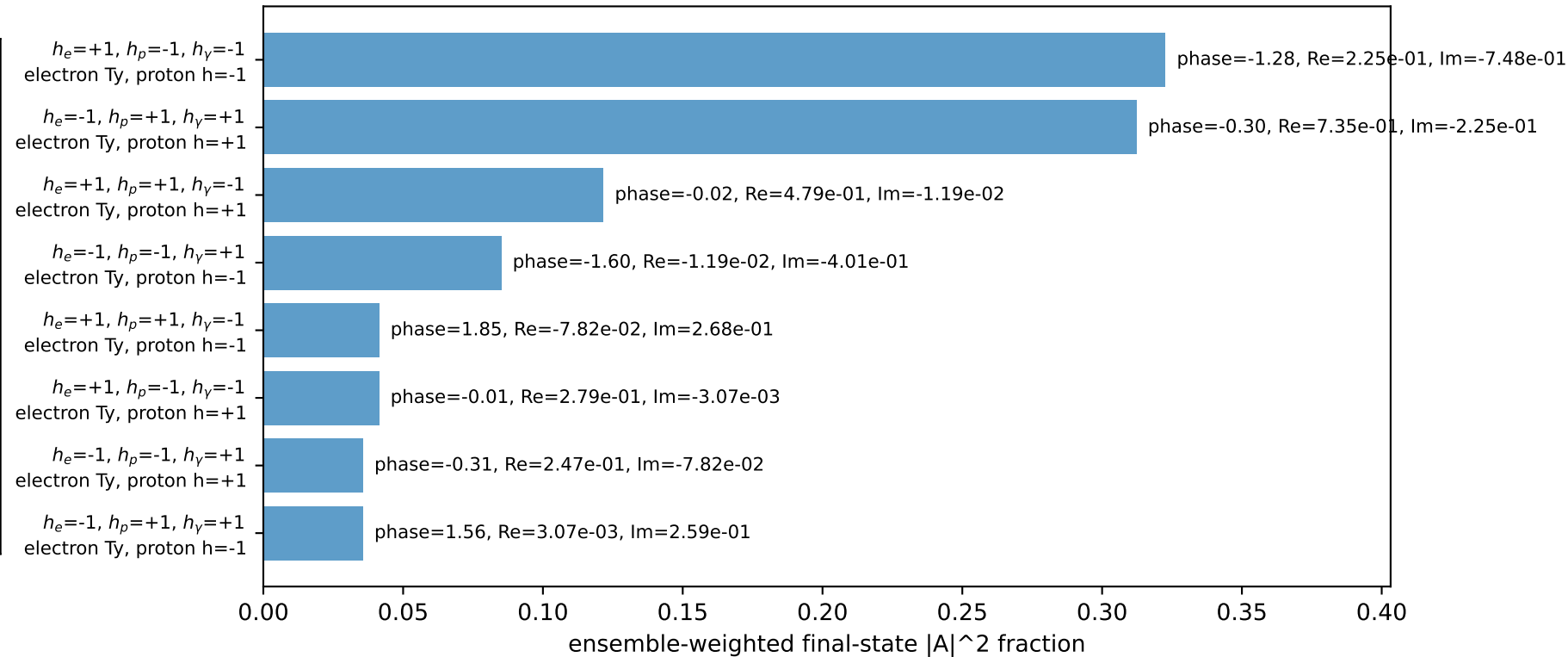
four-momenta $[E, p_x, p_y, p_z]$ GeV:

l $E= 2.2256$ $p_x= 2.1274$ $p_y= -0.6453$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -2.1274$ $p_y= 0.6453$ $p_z= 0.0000$
 l' $E= 1.2266$ $p_x= 0.0000$ $p_y= -1.2220$ $p_z= 0.0000$
 P' $E= 2.4119$ $p_x= 0.0000$ $p_y= 2.2220$ $p_z= 0.0000$
 q_Y $E= 1.0000$ $p_x= -0.0000$ $p_y= -1.0000$ $p_z= 0.0000$

Transverse plane

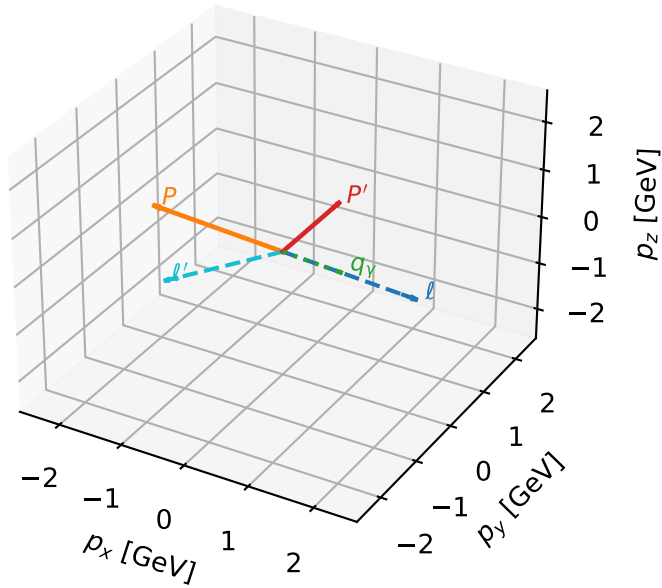


Leading initial-ensemble/final-state components (N=8, retained=99.5%)



Ty muon characteristic max P_W configuration

3D momenta

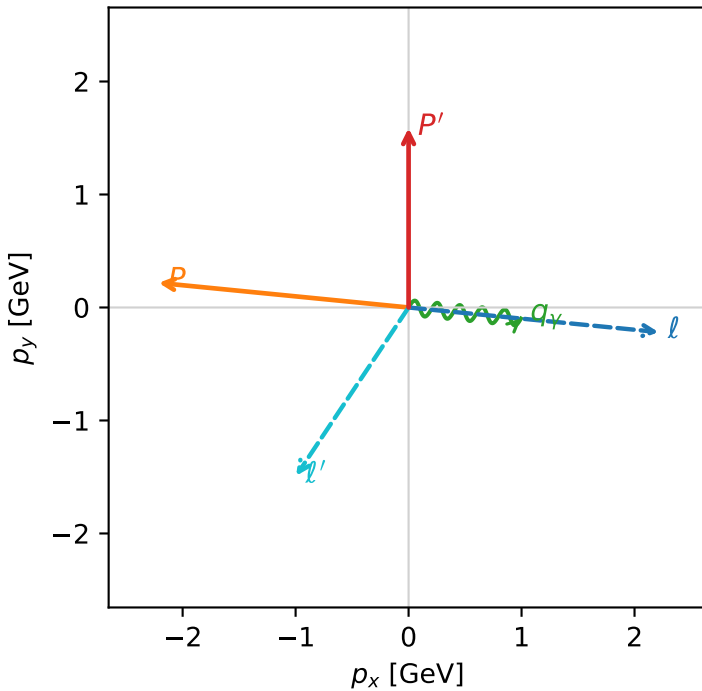


w_purity_mid_egamma_ty_lepton_region_1 $P_W=0.253092$
 region: mid_Egamma_Ty_lepton_region_1
 outgoing-state purity: 0.501237
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{y,\mu}, h_p)$

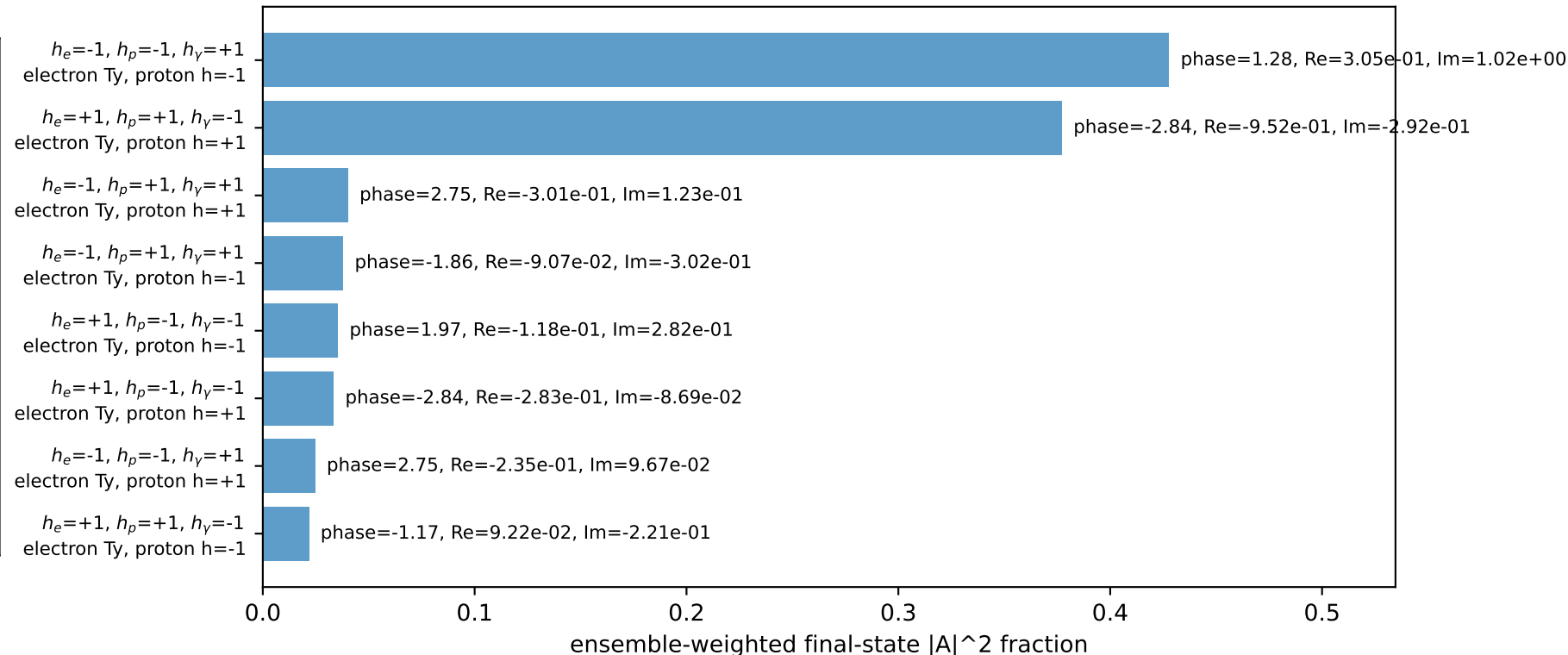
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.58758$
 $\theta_{in}=1.57079$, $\phi_l=6.18501$, $\phi_p=3.04342$
 $E_\gamma=1$, $\phi_\gamma=6.18501$
 $Q^2=11.7199$, $x_B=0.745585$, $t=-6.4471$, $W^2=4.87903$, $y=0.762145$
 $\theta(l', \gamma)=118.122$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= 2.2124$ $p_y= -0.2179$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -2.2124$ $p_y= 0.2179$ $p_z= 0.0000$
 l' $E= 1.7945$ $p_x= -0.9952$ $p_y= -1.4896$ $p_z= 0.0000$
 P' $E= 1.8440$ $p_x= 0.0000$ $p_y= 1.5876$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= 0.9952$ $p_y= -0.0980$ $p_z= 0.0000$

Transverse plane

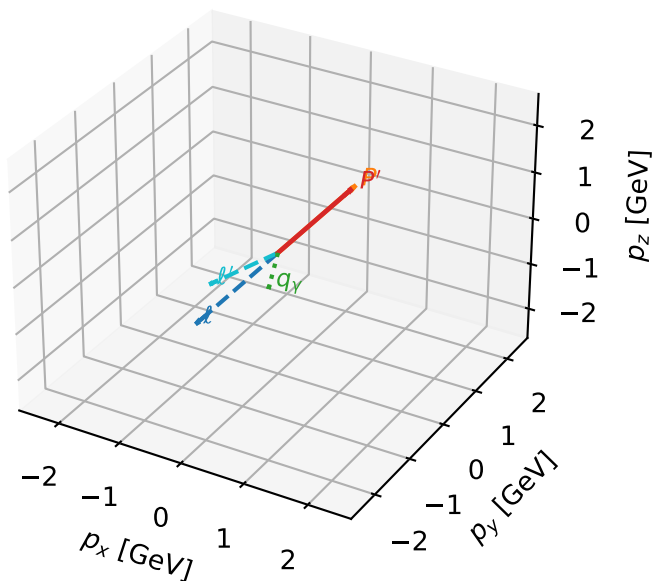


Leading initial-ensemble/final-state components (N=8, retained=99.8%)



Ty muon characteristic max P_W configuration

3D momenta

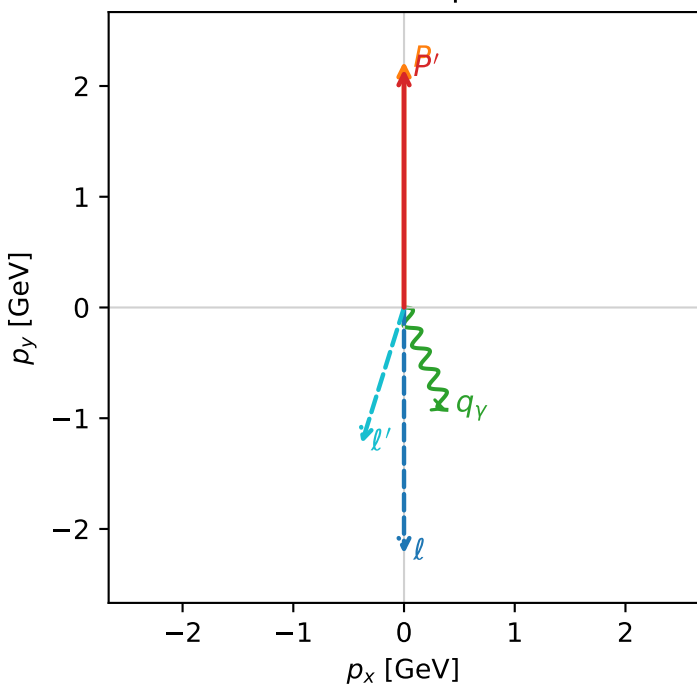


w_purity_mid_egamma_ty_lepton_region_2 P_W=0.224547
 region: mid_Egamma_Ty_lepton_region_2
 outgoing-state purity: 0.506942
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $\frac{1}{2} \sum_{h_p} \rho(T_{Y,\mu}, h_p)$

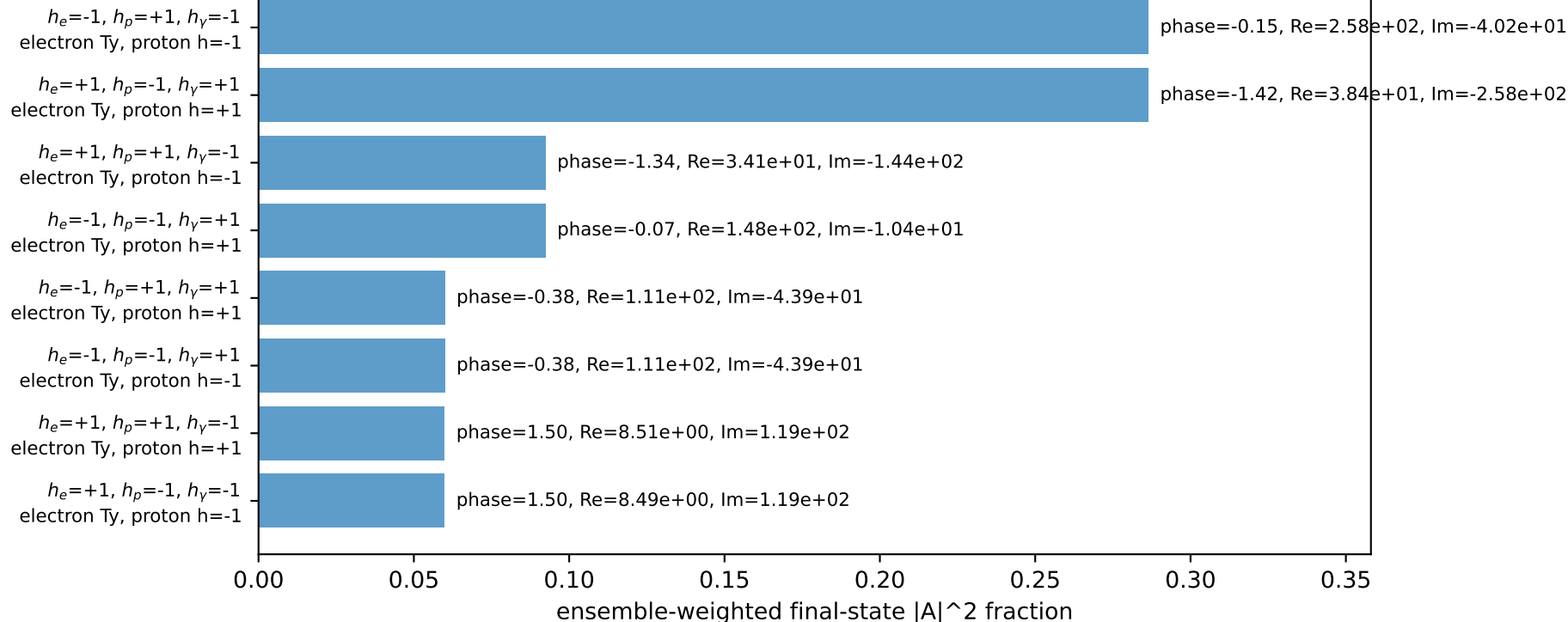
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=2.15215$
 $\theta_{in}=1.57079$, $\phi_l=4.71239$, $\phi_P=1.5708$
 $E_Y=1$, $\phi_Y=5.10509$
 $Q^2=0.262333$, $x_B=0.0293623$, $t=-0.000782118$, $W^2=9.55186$, $y=0.433184$
 $\theta(l', \gamma)=39.8051$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= -0.0000$ $p_y= -2.2231$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 0.0000$ $p_y= 2.2231$ $p_z= 0.0000$
 l' $E= 1.2908$ $p_x= -0.3827$ $p_y= -1.2283$ $p_z= 0.0000$
 P' $E= 2.3477$ $p_x= 0.0000$ $p_y= 2.1522$ $p_z= 0.0000$
 q_Y $E= 1.0000$ $p_x= 0.3827$ $p_y= -0.9239$ $p_z= 0.0000$

Transverse plane

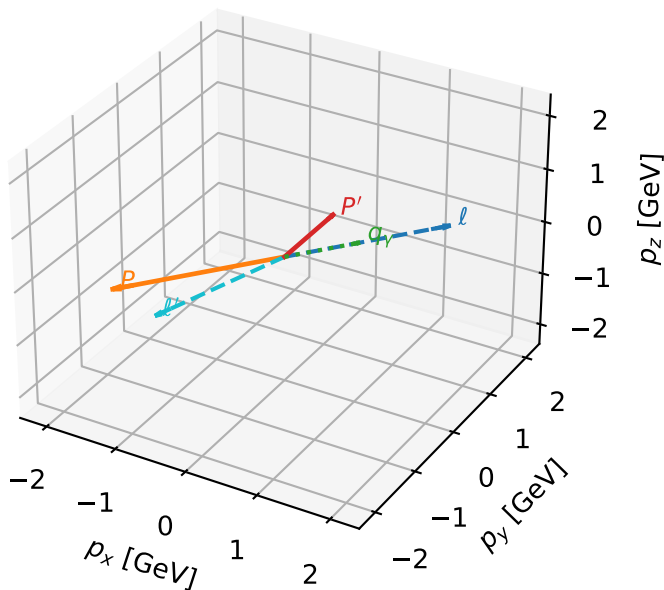


Leading initial-ensemble/final-state components (N=8, retained=99.7%)



Ty muon characteristic max P_W configuration

3D momenta

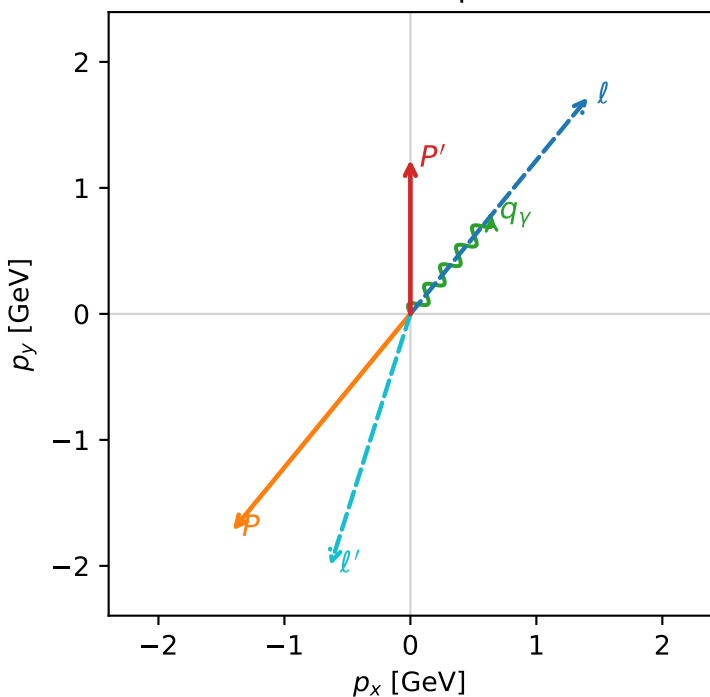


w_purity_mid_egamma_ty_lepton_region_3 P_W=0.183583
 region: mid_Egamma_Ty_lepton_region_3
 outgoing-state purity: 0.500082
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $\frac{1}{2} \sum_{h_p} \rho(T_{y,\mu}, h_p)$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.22309$
 $\theta_{in}=1.57079$, $\phi_l=0.883573$, $\phi_p=4.02517$
 $E_V=1$, $\phi_V=0.883573$
 $Q^2=17.9626$, $x_B=0.937779$, $t=-9.88236$, $W^2=2.07164$, $y=0.928705$
 $\theta(l', \gamma)=158.256$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= 1.4103$ $p_y= 1.7185$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -1.4103$ $p_y= -1.7185$ $p_z= 0.0000$
 l' $E= 2.0972$ $p_x= -0.6344$ $p_y= -1.9961$ $p_z= 0.0000$
 P' $E= 1.5414$ $p_x= 0.0000$ $p_y= 1.2231$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= 0.6344$ $p_y= 0.7730$ $p_z= 0.0000$

Transverse plane



$h_e=-1, h_p=-1, h_\gamma=+1$
 electron Ty, proton h=-1

$h_e=+1, h_p=+1, h_\gamma=-1$
 electron Ty, proton h=+1

Leading initial-ensemble/final-state components (N=2, retained=98.0%)

